EEE-4484

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Section: 1A

Dept. CSE

Expt. Name: Study of ~~RTC NOR Gate and TTL~~ NA

Operational Amplifier as Triangular Wave Generator

and Astable Generator

Expt. No. 03

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Objective:

- Understand the fundamental functions and applications of an operational amplifier,
- Explore how to design and analyze a bipolar triangular wave generator using an op-amp.
- Learn the working principle and construction of a unipolar triangle wave generator with an op-amp.
- Develop and test an astable multivibrator circuit using an op-amp to generate continuous waveform without external triggering

Equipment:

- DC power supply
- Signal / function generator
- Breadboard
- Resistors
- Diode
- Capacitors
- Op-amp (LM741)
- Oscilloscope
- Connecting wires

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Bipolar Triangle Wave Generator using Op-amp:

Circuit Diagram:

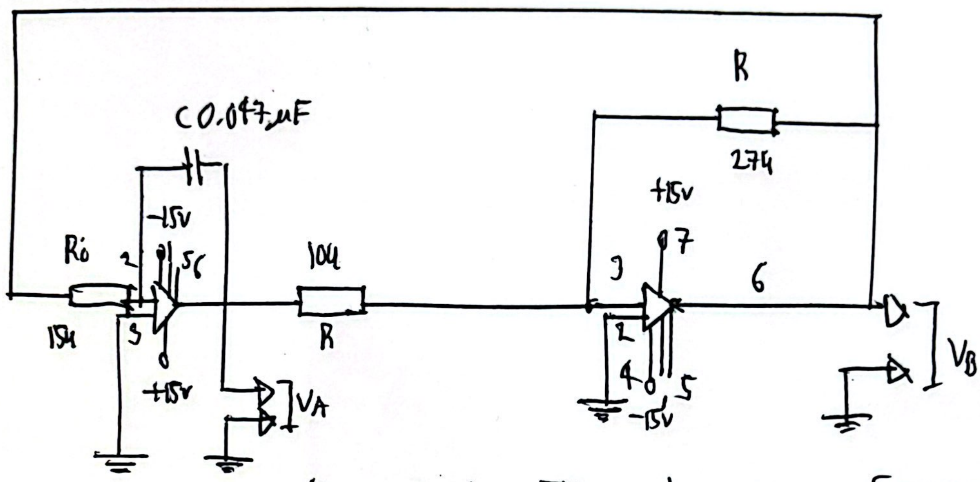
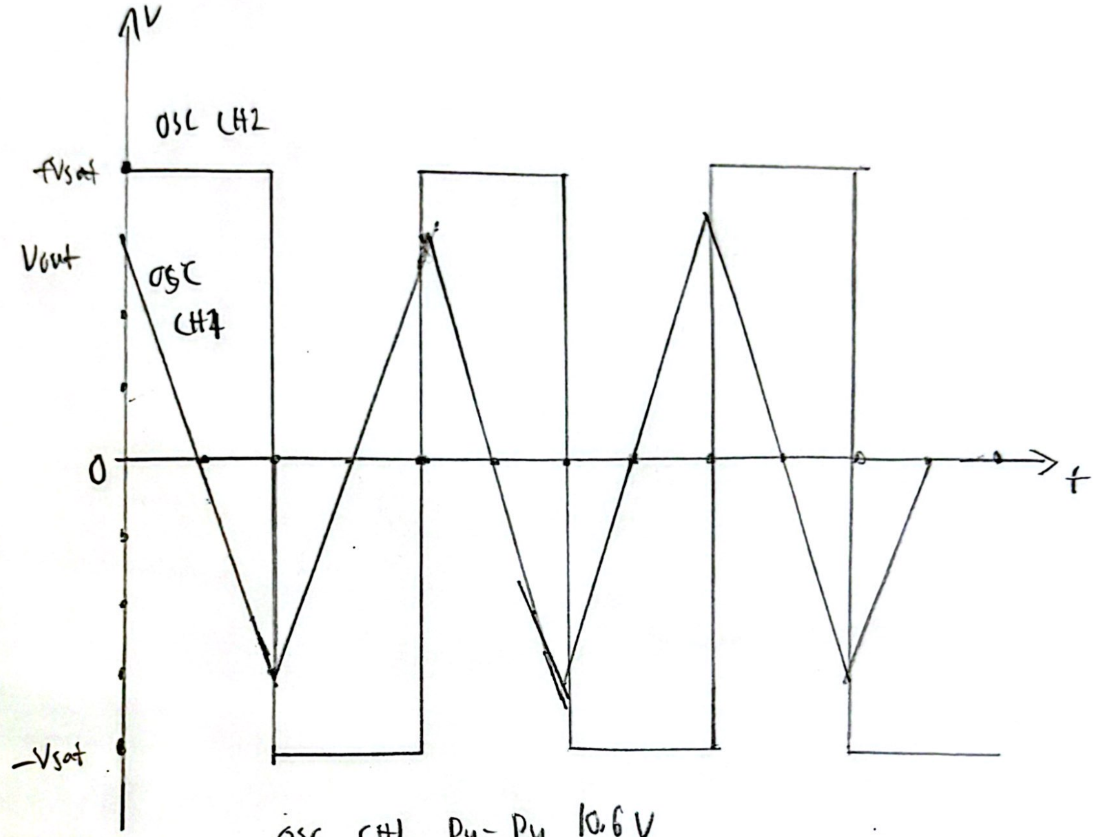


Fig: Bipolar Triangular Wave Generator

Justification:

The comparator, O2 switches its output between the positive and negative saturation voltages (+15V and -15V) depending on the input from the integrator, O1. This output behaves like square waves and is fed back to the input of the integrator through resistor R. The integrator receives the square wave signal at its input and produces a linearly increasing or decreasing voltage across capacitor C, forming a triangular waveform.

The output voltage is fed back to the non-inverting terminal of the comparator through resistor network Ri, pR.



OSC CH1 Pk-Pk 10.6 V

OSC CH2 Pk-Pk 27.6V

Unipolar Triangle Wave Generator using OP-Amp:

Circuit Diagram:

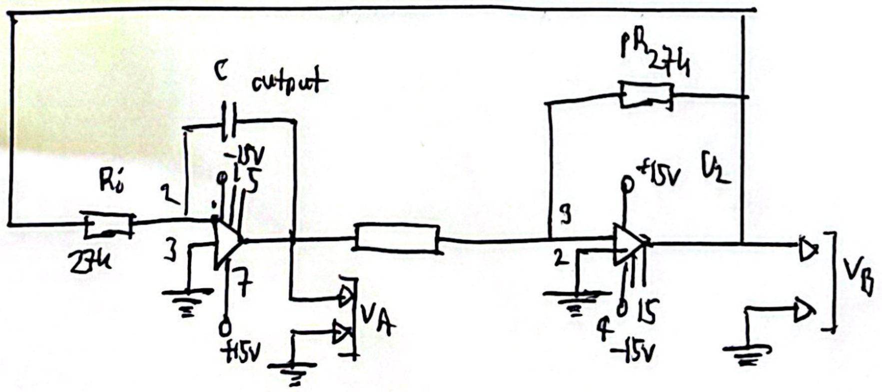


Fig: Unipolar Triangle Wave generator

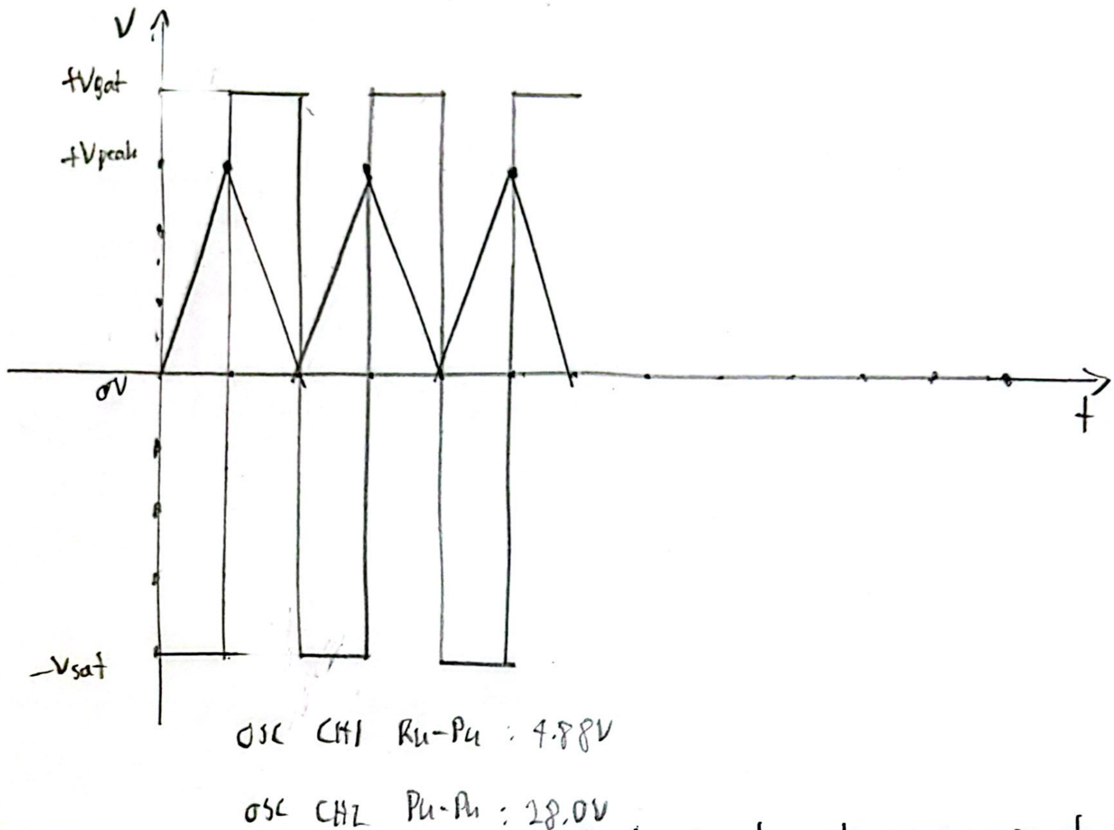


Fig. ~~Fig~~: Graph. of Unipolar triangle wave generator

__Justification:__

Here, ~~U2~~ U2 operates as a comparator producing a wave of square shape while U1 acts as an integrator to generate the triangular waveform. The diode in the feedback path ensured that the comparator outside stayed within a fixed range instead of oscillating between ±V as in bipolar version. This effectively shifts the triangle's waveform to the positive region, achieving unipolar behavior.

Astable Multivibrator using Op-Amp:

Circuit Diagram:

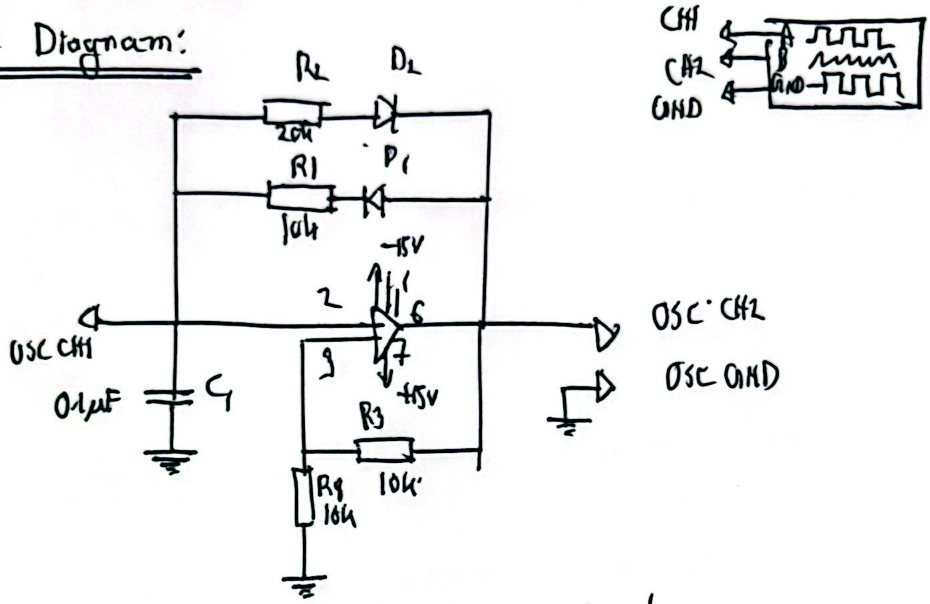
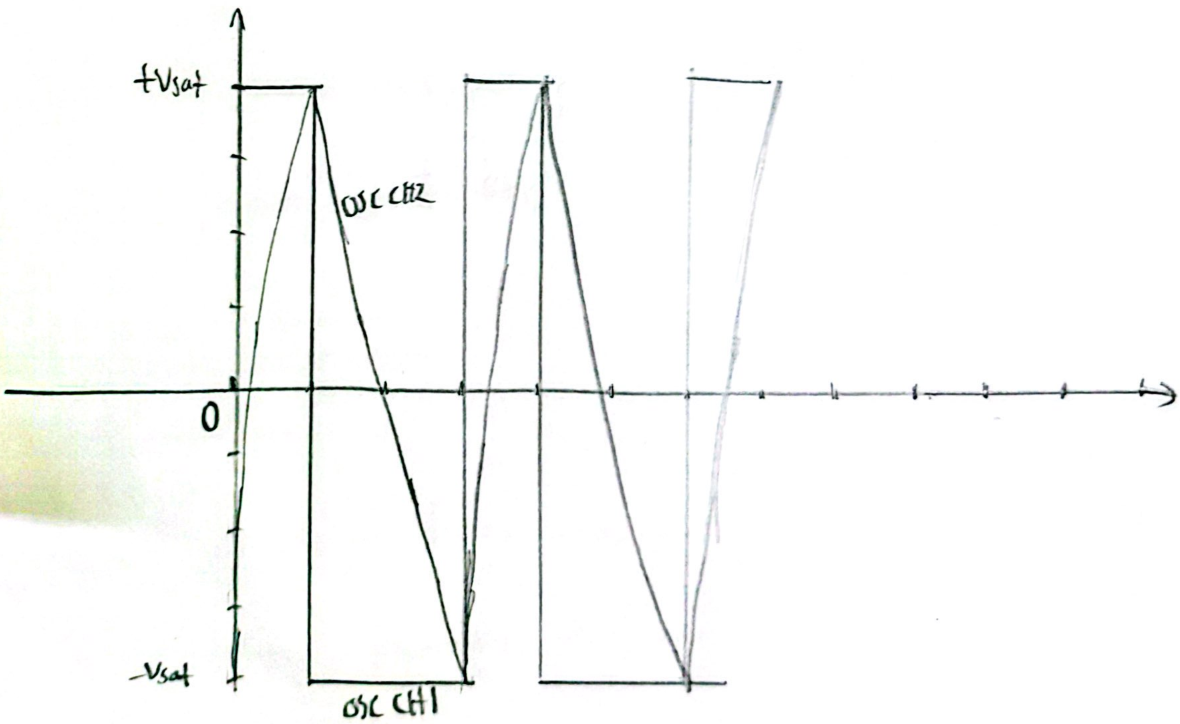


Fig: Astable Multivibrator



OSC CH1 Pk-Pk : 28.8V

OSC CH2 Pk-Pk : 14.2V

Fig. Astable Multivibrator Graph.

Justification:

The circuit was configured with positive feedback and diode steering to generate a continuous square wave without any external triggering. The RC network (C1, R1, R2) determines the charging / discharging cycle of the capacitor, ~~which divides~~ while diode D1, D2 direct current flow to control timing during each half cycle. As the capacitor charges and discharges, the Op-amp output toggles between saturation levels producing an oscillating output signal. The waveform observed on the oscilloscope confirmed stable, periodic wave generation,

Answer to Report Questions;

Answer to Question 1:-

Both circuits in figure 3(c) and fig 3(d) are triangular wave generators having primarily difference in the nature of output waveform;

— fig 3(c) generates bipolar triangle wave, where the waveform oscillates symmetrically around 0V, The comparator output the generator .

swings between positive and negative saturation, feeding the integrator to produce a centered triangle wave.

- Fig 3 (d) produces a unipolar triangle wave, where the waveform remains entirely above 0V. This is achieved by adding a diode at the output of the comparator which clamps negative part of the waveform, allowing only positive voltages to reach the integrator. As a result, triangle wave shifts upwards and become unipolar.

Answer to Question 2:

The circuit in Fig.3 (e) is an astable vibrator. It functions as a self-generating square wave generator with no external input. The key components include a capacitor (C1), resistors (R1, R2, R3, R4) and diode (D1, D2). The capacitor charges and discharges alternatively through R1; R2 while the op-amp provides positive feedback via voltage dividers and the diode network. When the capacitor voltage reaches the swinging

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threshold determined by the feedback path, the op-amp output toggles between high and low saturation levels. This transition reverses the charging direction of the capacitor, resulting in continuous square wave generation at the output. D1, D2 control the symmetry and timing of the waveform by setting precise threshold.

Answer to Question 3:

The experiments constructed demonstrated the use of Op-amps in waveform generation. The bipolar and unipolar wave generator circuits showcased how feedback loops between a comparator and an integrator can produce continuous oscillations. The inclusion of diode clamping in the unipolar generator allowed control over the voltage range of the output. The astable multivibrator further illustrated the principle of self-triggering oscillation using positive feedback and RC timing. Across all circuits, the use of simple Op-Amp configurations enabled the generation of clean and predictable waveforms.

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